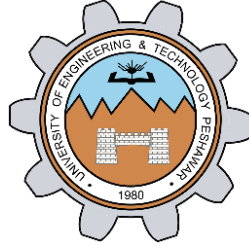


Introduction to xilinx ise and spartan 6 board

LAB # 04



Spring 2024

CSE-308L Digital System Design Lab

Submitted by: **Ali Asghar**

Registration No.: **21PWCSE2059**

Class Section: **C**

“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

Dr. Asif Ali Khan

Date:

22nd March 2024

Department of Computer Systems Engineering
University of Engineering and Technology, Peshawar

Objectives:

This lab will enable students to:

- Introduction to FPGA
- Introduction to Xilinx ISE

Software used:

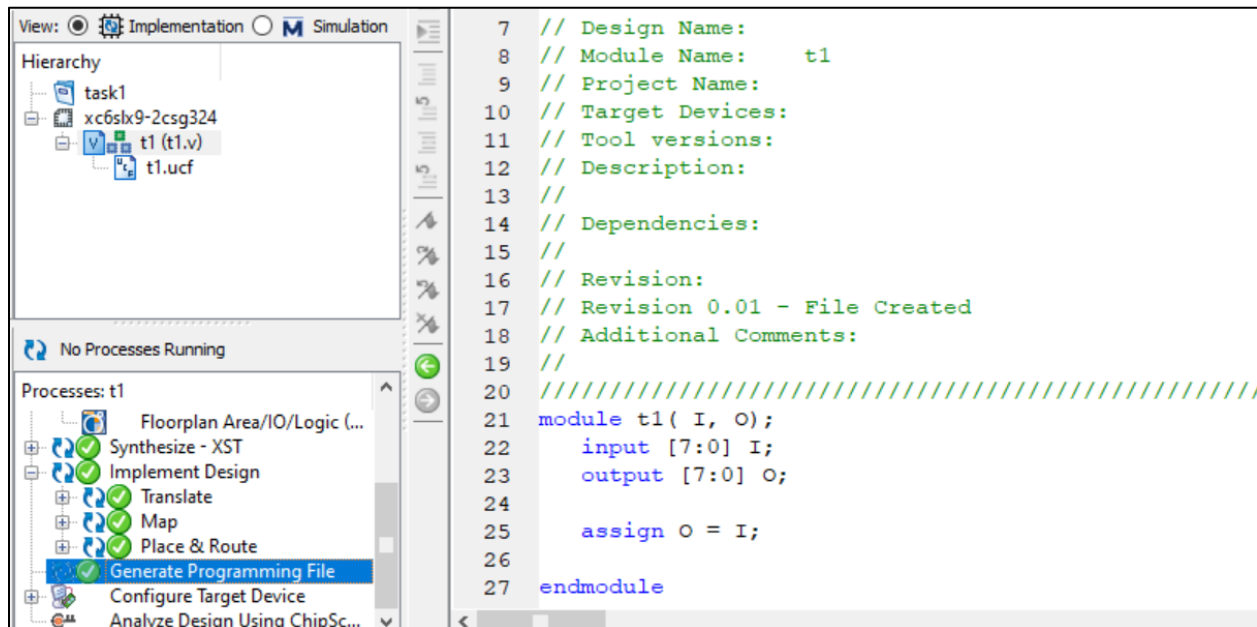
- Xilinx ISE

Task 1:

Develop a program to control on Board LED using On board available switch.

Code:

DUT:



ucf file:

```
1  # PlanAhead Generated physical constraints
2
3  NET "I[0]" LOC = F17;
4  NET "I[1]" LOC = F18;
5  NET "I[2]" LOC = E16;
6  NET "I[3]" LOC = E18;
7  NET "I[4]" LOC = D18;
8  NET "I[5]" LOC = D17;
9  NET "I[6]" LOC = C18;
10 NET "I[7]" LOC = C17;
11
12 NET "O[0]" LOC = P15;
13 NET "O[1]" LOC = P16;
14 NET "O[2]" LOC = N15;
15 NET "O[3]" LOC = N16;
16 NET "O[4]" LOC = U17;
17 NET "O[5]" LOC = U18;
18 NET "O[6]" LOC = T17;
19 NET "O[7]" LOC = T18;
20
21 # PlanAhead Generated IO constraints
22
23 NET "I[0]" PULLUP;
24 NET "I[1]" PULLUP;
25 NET "I[2]" PULLUP;
```

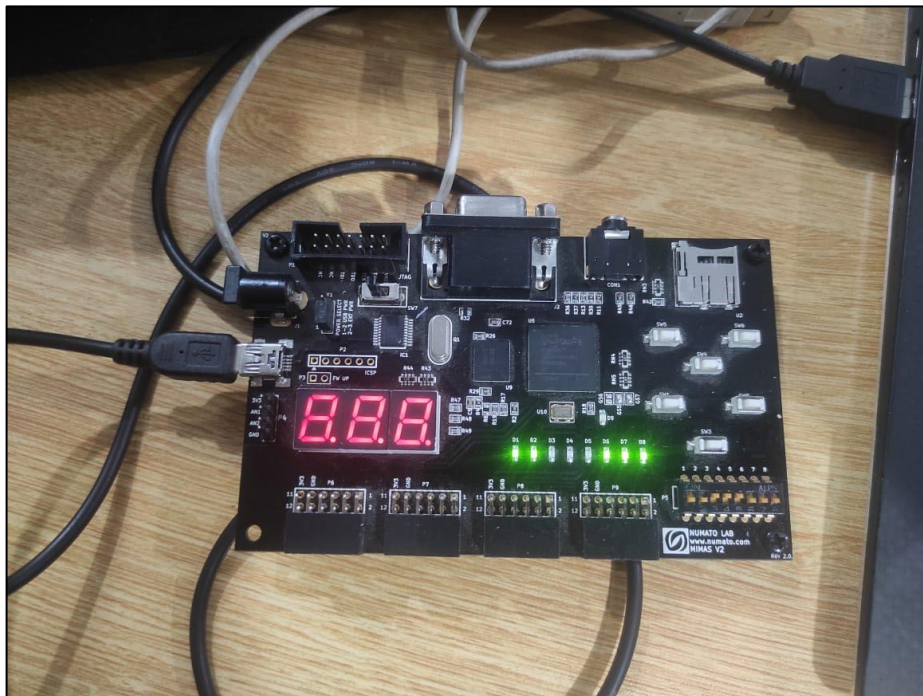
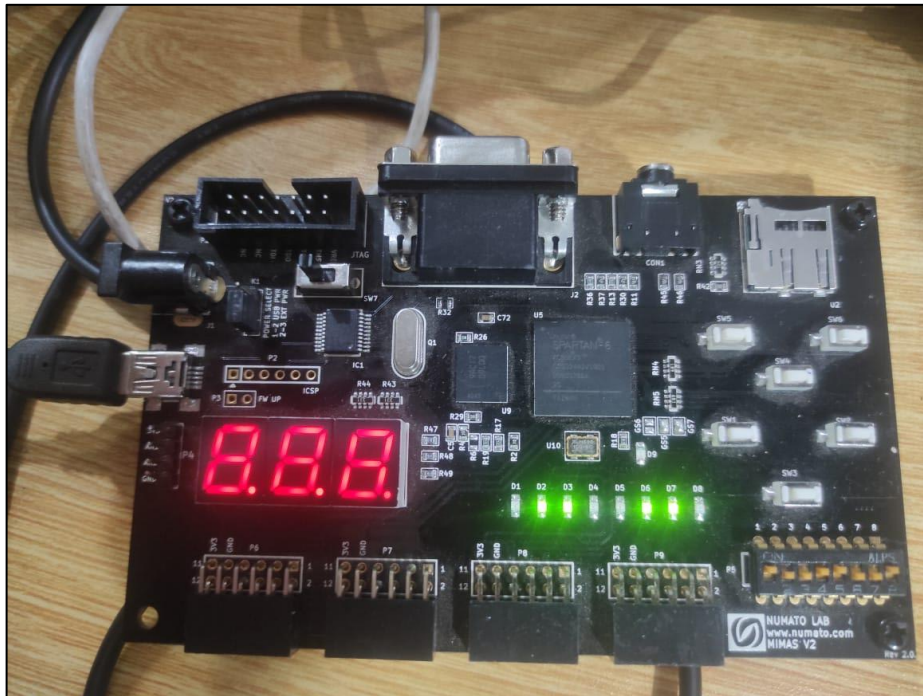
Design Summary (Programming File Generated) t1.v t1.ucf

```
23 NET "I[0]" PULLUP;
24 NET "I[1]" PULLUP;
25 NET "I[2]" PULLUP;
26 NET "I[3]" PULLUP;
27 NET "I[4]" PULLUP;
28 NET "I[5]" PULLUP;
29 NET "I[6]" PULLUP;
30 NET "I[7]" PULLUP;
31
32 NET "O[0]" SLEW = FAST;
33 NET "O[1]" SLEW = FAST;
34 NET "O[2]" SLEW = FAST;
35 NET "O[3]" SLEW = FAST;
36 NET "O[4]" SLEW = FAST;
37 NET "O[5]" SLEW = FAST;
38 NET "O[6]" SLEW = FAST;
39 NET "O[7]" SLEW = FAST;
```

Design Summary (Programming File Generated) t1.v t1.ucf

Output:

All the dip switches represent I while LEDs represent O.

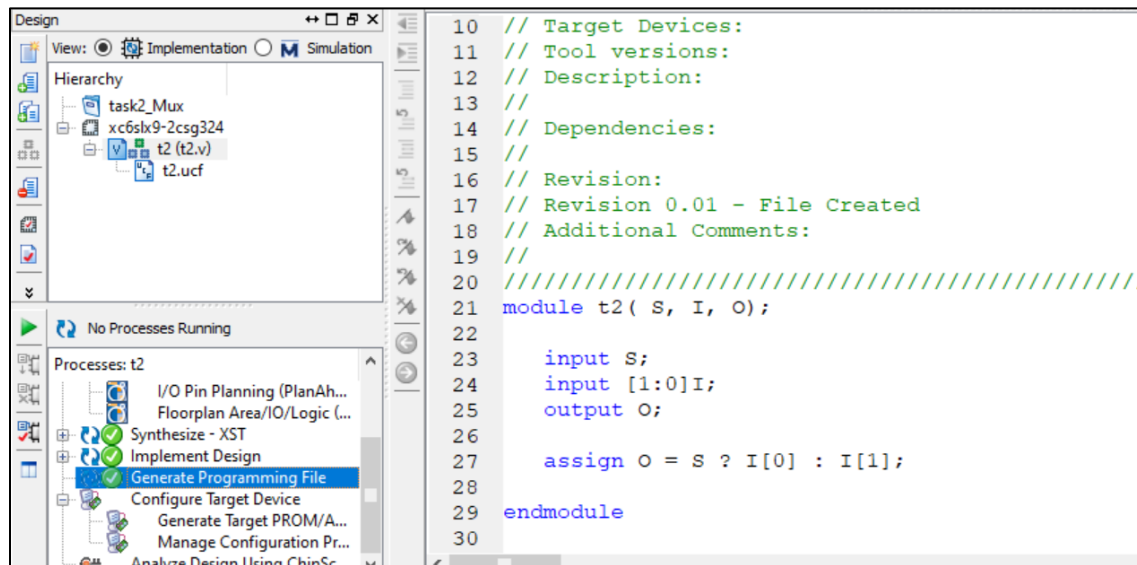


Task 2:

Develop a program that implements a 2x1 multiplexer on the board. Connect the inputs to switches and output to led.

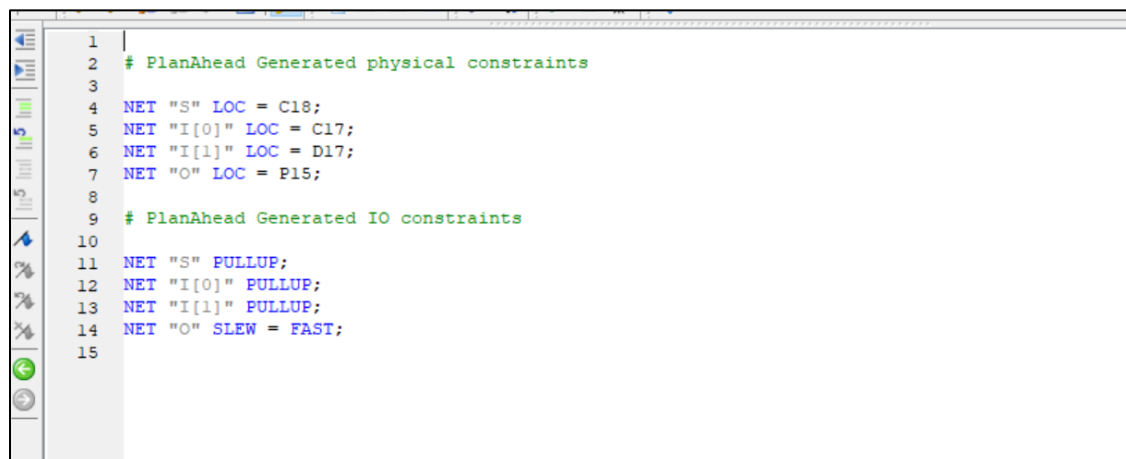
Code:

DUT:



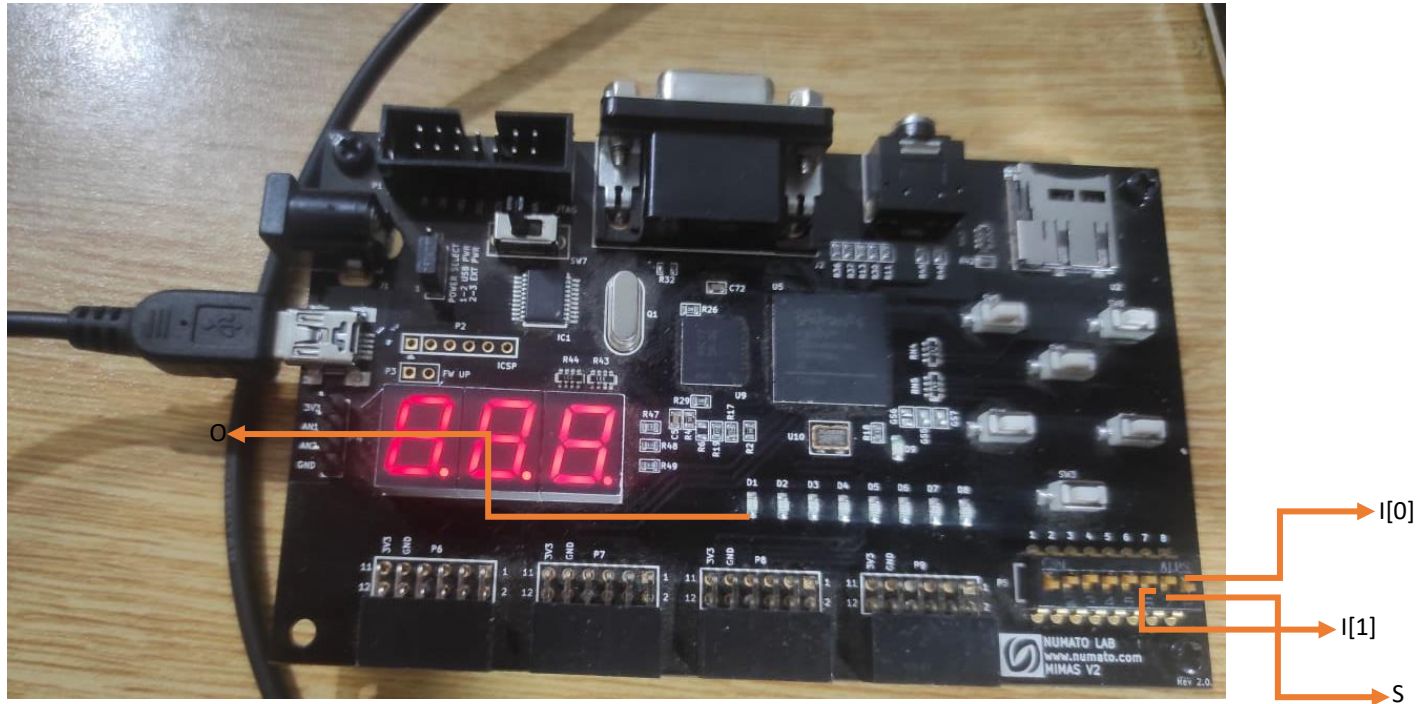
```
10 // Target Devices:
11 // Tool versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////
21 module t2( S, I, O);
22
23     input S;
24     input [1:0]I;
25     output O;
26
27     assign O = S ? I[0] : I[1];
28
29 endmodule
30
```

ucf file:



```
1 |
2 # PlanAhead Generated physical constraints
3
4 NET "S" LOC = C18;
5 NET "I[0]" LOC = C17;
6 NET "I[1]" LOC = D17;
7 NET "O" LOC = P15;
8
9 # PlanAhead Generated IO constraints
10
11 NET "S" PULLUP;
12 NET "I[0]" PULLUP;
13 NET "I[1]" PULLUP;
14 NET "O" SLEW = FAST;
15
```

Output:

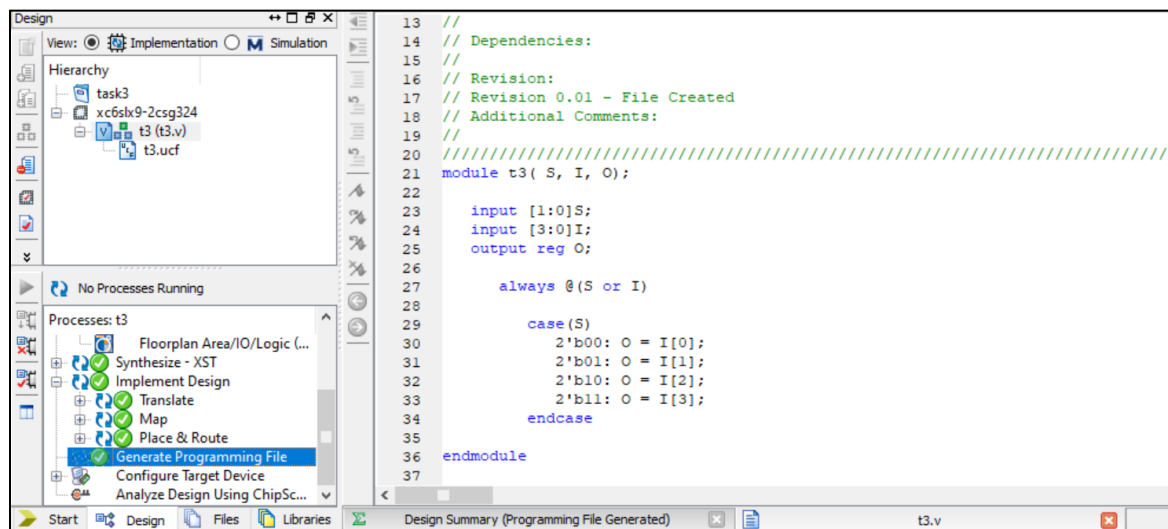


Task 3:

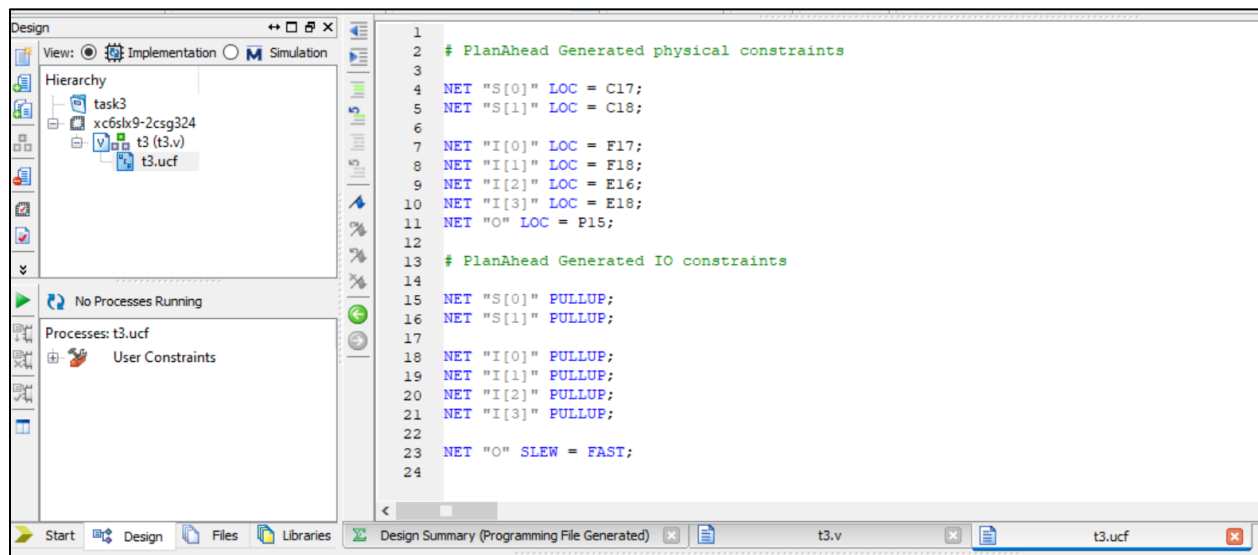
Develop a program that implements a 4x1 multiplexer on the board. Connect the inputs to switches and output to led

Code:

DUT:



ucf file:



Output:

